

Computational Logic

Normal forms and satisfiability

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We have define:

- Notion of satisfiability, semantical consequence and semantical equivalence
- Truth (boolean) functions. Complete sets of connectives

Normal forms

Normal forms

We now see that we can transform formulas into semantically equivalent formulas to obtain formulas of **special forms** and that allow us to decide more efficiently the satisfiability or validity of the original formulas... some of these **normal forms** exist for any formula, while others **only** for certain classes of formulas...

Negative normal form

A **literal** is a propositional variable p or its negation, $\neg p$.

A formula is said to be in **negative normal form** if negation ($\neg$) occurs only in literals.

Proposition

Any formula containing only the connectives $\wedge$, $\vee$, and $\neg$ is semantically equivalent to a negative normal form formula.

Proof.

Just use DeMorgan's Laws and eliminate double negations. □

Negative normal form

Example

$$\neg((p \vee q) \wedge \neg p)$$

$$(DeMorgan) \Leftrightarrow \neg(p \vee q) \vee \neg\neg p$$

$$(DeMorgan) \Leftrightarrow (\neg p \wedge \neg q) \vee \neg\neg p$$

$$(Double\ Negation) \Leftrightarrow (\neg p \wedge \neg q) \vee p$$

Disjunctive normal forms

A formula is said to be in **disjunctive normal form** if it is of the form:

$$(\alpha_{11} \wedge \cdots \wedge \alpha_{1k_1}) \vee \cdots \vee (\alpha_{n1} \wedge \cdots \wedge \alpha_{nk_n})$$

where each α_{ij} is a literal.

Examples of disjunctive normal forms:

$$\begin{aligned} & p \wedge q \\ & p \vee \neg p \\ & (p \wedge q \wedge \neg r) \vee (\neg p \wedge r) \end{aligned}$$

Disjunctive normal forms

Lemma

For any function $f : \{\mathbf{V}, \mathbf{F}\}^n \longrightarrow \{\mathbf{V}, \mathbf{F}\}$, there is a formula ϕ with n propositional variables in disjunctive normal form, such that $F_\phi = f$.

x_1	x_2	$F(x_1, x_2)$
F	F	F
F	V	F
V	F	F
V	V	V

$$p_1 \wedge p_2$$

Proof

If f is **F**, for all argument values, then $\phi = p_1 \wedge \neg p_1$.

Otherwise, for each assignment v let

$$I_i^v = \begin{cases} p_i & \text{if } v(x_i) = \mathbf{V} \\ \neg p_i & \text{if } v(x_i) = \mathbf{F} \end{cases}$$

What is $v(I_i^v)$? And then

$$\phi_v = I_1^v \wedge \cdots \wedge I_n^v$$

Note that $v(\phi_v) = \mathbf{V}$. So just consider

$$\phi = \bigvee_{f(v(p_1), \dots, v(p_n)) = \mathbf{V}} \phi_v$$

Example

For the following truth function:

x_1	x_2	x_3	$f(x_1, x_2, x_3)$
V	V	V	V
V	V	F	V
V	F	V	V
V	F	F	F
F	V	V	V
F	V	F	F
F	F	V	F
F	F	F	F

a formula in disjunctive normal form is:

$$(p_1 \wedge p_2 \wedge p_3) \vee (p_1 \wedge p_2 \wedge \neg p_3) \vee (p_1 \wedge \neg p_2 \wedge p_3) \vee (\neg p_1 \wedge p_2 \wedge p_3)$$

Disjunctive normal forms

Any formula is semantically equivalent to a formula in disjunctive normal form.

Proof.

Given a formula it is possible to transform it into a semantically equivalent formula in disjunctive normal form, considering the following steps:

1. get a formula with just the connectives $\wedge$, $\vee$ and $\neg$
2. get a formula in negative normal form
3. apply distributivity:

$$(\phi \vee \psi) \wedge \theta \Leftrightarrow (\phi \wedge \theta) \vee (\psi \wedge \theta)$$



Example

Determine a disjunctive normal form for:

$$(p \vee r) \leftrightarrow (q \wedge \neg p)$$

$$(p \vee r) \leftrightarrow (q \wedge \neg p)$$

$$\Leftrightarrow ((p \vee r) \rightarrow (q \wedge \neg p)) \wedge ((q \wedge \neg p) \rightarrow (p \vee r))$$

$$\Leftrightarrow (\neg(p \vee r) \vee (q \wedge \neg p)) \wedge (\neg(q \wedge \neg p) \vee (p \vee r))$$

$$\Leftrightarrow ((\neg p \wedge \neg r) \vee (q \wedge \neg p)) \wedge ((\neg q \vee p) \vee (p \vee r))$$

$$\Leftrightarrow (((\neg p \wedge \neg r) \vee (q \wedge \neg p)) \wedge (\neg q \vee p)) \vee (((\neg p \wedge \neg r) \vee (q \wedge \neg p)) \wedge (p \vee r))$$

$$\Leftrightarrow (\neg p \wedge \neg r \wedge (\neg q \vee p)) \vee (q \wedge \neg p \wedge (\neg q \vee p)) \\ \vee (\neg p \wedge \neg r \wedge (p \vee r)) \vee (q \wedge \neg p \wedge (p \vee r))$$

$$\Leftrightarrow (\neg p \wedge \neg r \wedge \neg q) \vee (\neg p \wedge \neg r \wedge p) \vee (q \wedge \neg p \wedge \neg q) \vee (q \wedge \neg p \wedge p) \\ \vee (\neg p \wedge \neg r \wedge p) \vee (\neg p \wedge \neg r \wedge r) \vee (q \wedge \neg p \wedge p) \vee (q \wedge \neg p \wedge r)$$

Satisfiability of formulas in DNF

A conjunction of literals $l_1 \wedge \dots \wedge l_n$ is satisfiable if and only if for all $1 \leq i, j \leq n$, l_i is not $\neg l_j$.

Are these satisfiable?

$p \wedge \neg q \wedge \neg r \wedge q$ No

$\neg p \wedge q \wedge \neg r \wedge \neg s$ Yes

Corollary

A formula ϕ in disjunctive normal form is satisfiable if and only if any of its conjunctions of literals is.

We thus obtain a method of determining whether a formula is **satisfiable**.

Example

Determine whether

$$(p \vee r) \leftrightarrow (q \wedge \neg p)$$

is satisfiable...

$$\begin{aligned}(p \vee r) &\leftrightarrow (q \wedge \neg p) \Leftrightarrow \\ (\neg p \wedge \neg r \wedge \neg q) \vee (\neg p \wedge \neg r \wedge p) \vee (q \wedge \neg p \wedge \neg q) \vee \\ (q \wedge \neg p \wedge p) \vee (\neg p \wedge \neg r \wedge p) \vee (\neg p \wedge \neg r \wedge r) \\ &\vee (q \wedge \neg p \wedge p) \vee (q \wedge \neg p \wedge r)\end{aligned}$$

$$\begin{aligned}(p \vee r) &\leftrightarrow (q \wedge \neg p) \Leftrightarrow \\ (\neg p \wedge \neg r \wedge \neg q) \vee (\neg p \wedge \neg r \wedge p) \vee (q \wedge \neg p \wedge \neg q) \vee \\ (q \wedge \neg p \wedge p) \vee (\neg p \wedge \neg r \wedge p) \vee (\neg p \wedge \neg r \wedge r) \\ &\vee (q \wedge \neg p \wedge p) \vee (q \wedge \neg p \wedge r)\end{aligned}$$

Yes

Conjunctive Normal Form

A formula is said to be in **conjunctive normal form** if it is of the form:

$$(\alpha_{11} \vee \cdots \vee \alpha_{1k_1}) \wedge \cdots \wedge (\alpha_{n1} \vee \cdots \vee \alpha_{nk_n})$$

where each α_{ij} is a literal.

By duality we have:

A disjunction of literals $l_1 \vee \cdots \vee l_n$ is a **tautology** if and only if for some $1 \leq i, j \leq n$, l_i is $\neg l_j$.

Conjunctive Normal Form

So it is easy to determine whether a formula in conjunctive normal form is a **tautology**: check that all disjunctions are tautologies.

But how to get a formula in conjunctive normal form?

1. if we have the truth table, by a method dual to that of the disjunctive normal form: that is, choose the lines that correspond to **F** consider for each one the disjunction of literals such that $p_i = \mathbf{V}$ put $\neg p_i$ and if $p_i = \mathbf{F}$ put p_i ; and finally take the conjunction of these disjunctions.
2. adapt the given method to the disjunctive normal form, using the distributivity for the conjunction:

$$(\phi \wedge \psi) \vee \theta \Leftrightarrow (\phi \vee \theta) \wedge (\psi \vee \theta)$$

Exercise

Obtain a formula in conjunctive normal form corresponding to the truth table:

x_1	x_2	x_3	$f(x_1, x_2, x_3)$
V	V	V	V
V	V	F	V
V	F	V	V
V	F	F	F
F	V	V	V
F	V	F	F
F	F	V	F
F	F	F	F

$$(\neg p_1 \vee p_2 \vee p_3) \wedge (p_1 \vee \neg p_2 \vee p_3) \wedge (p_1 \vee p_2 \vee \neg p_3) \wedge (p_1 \vee p_2 \vee p_3)$$

Exercise

Determine a conjunctive normal form for the formula below and check if it is a tautology

$$(p \vee r) \leftrightarrow (q \wedge \neg p)$$

$$(p \vee r) \leftrightarrow (q \wedge \neg p)$$

$$\Leftrightarrow ((p \vee r) \rightarrow (q \wedge \neg p)) \wedge ((q \wedge \neg p) \rightarrow (p \vee r))$$

$$\Leftrightarrow (\neg(p \vee r) \vee (q \wedge \neg p)) \wedge (\neg(q \wedge \neg p) \vee (p \vee r))$$

$$\Leftrightarrow ((\neg p \wedge \neg r) \vee (q \wedge \neg p)) \wedge ((\neg q \vee p) \vee (p \vee r))$$

$$\Leftrightarrow ((\neg p \wedge \neg r) \vee q) \wedge ((\neg p \wedge \neg r) \vee \neg p) \wedge (\neg q \vee p \vee p \vee r)$$

$$\Leftrightarrow (\neg p \vee q) \wedge (\neg r \vee q) \wedge (\neg p \vee \neg p) \wedge (\neg r \vee \neg p) \wedge (\neg q \vee p \vee p \vee r)$$

No

Horn's formulas

A *Horn's formula* is a formula in conjunctive normal form in which in each disjunction (clause) there is at most one positive literal.

$$p \wedge \neg q \wedge (q \vee \neg p)$$

$$(\neg p \vee \neg q \vee \neg s \vee p) \wedge (\neg q \vee \neg r \vee p) \wedge (\neg p \vee \neg s \vee s)$$

$$(\neg p \vee \neg q \vee \neg s) \wedge (\neg q \vee \neg r \vee p) \wedge s$$

In a Horn formula, the disjunctions $\neg p_1 \vee \dots \vee \neg p_n \vee p$ can also be written as:

$$(p_1 \wedge \dots \wedge p_n) \rightarrow p$$

or if p does not exist (or is **F**):

$$(p_1 \wedge \dots \wedge p_n) \rightarrow \mathbf{F}$$

or if the p_i do not exist:

$$\mathbf{V} \rightarrow p$$

Not all formulas have an equivalent Horn formula!

But...

To determine whether a Horn formula of propositional logic is **satisfiable** we can use an efficient algorithm!

Satisfiability of a Horn formula

Consider the formula $p \wedge \neg q \wedge (q \vee \neg p)$:

- start by placing in a line the propositional variables that occur in the formula and the formula. Ex:

$$\frac{p \mid q \parallel p \wedge \neg q \wedge (q \vee \neg p)}{\quad}$$

- if any of the propositional variables is one of the elements of the conjunction assign the value **V** to that variable (why?).

Ex:

$$\frac{p \mid q \parallel p \wedge \neg q \wedge (q \vee \neg p)}{\mathbf{V} \mid \quad \parallel}$$

Satisfiability of a Horn formula

- With this information, fill in the table as if you were constructing the truth table (for this line), analyzing each disjunction to determine if, for it to be true, more values can be determined for the propositional variables:

p	q	$p \wedge \neg q \wedge (q \vee \neg p)$					
V		F					

- In this case, q must be **V** and so this can be added:

p	q	$p \wedge \neg q \wedge (q \vee \neg p)$					
V	V	F					

Satisfiability of a Horn formula

- And repeating this step again, we get:

p	q	$p \wedge \neg q \wedge (q \vee \neg p)$				
V	V	F				

Continue until nothing more can be added.

- if in the previous step **F** is assigned to one of the elements of the conjunction, the formula also has the value **F** and is not satisfiable. Otherwise we can assign the value **V** to the formula if we assign **F** to the remaining propositional variables.

In the example we are considering, the formula has the value **F** and therefore is not satisfiable.